

Design & Simulation of JamSail's ADCS

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ABSTRACT

GNSS technology is integral to critical infrastructure, and faces growing challenges due to signal interference, with organizations like EASA and Airbus highlighting concern on its impact. The research group at University of Nottingham's, NottSpace, is developing its first CubeSat with the aim to generate global GNSS interference heatmaps. The mission relies on a robust Attitude Determination Control System (ADCS) for accurate data collection and directional control. This project focuses on developing a simulation environment to verify and validate the ADCS, targeting an attitude estimation accuracy within 10° , pointing accuracy within 20° , and detumbling from 90° to 0° in 60s. An Extended Kalman Filter was implemented using magnetometer, gyro, and Sun sensor data, alongside quaternion-based control. The N-Body Simulator, developed in accordance with ECSS standards, was used to test ADCS performance. Results show attitude errors under 3° when the Sun sensor is active, with control accuracy meeting mission requirements. The data collected by JamSail will support the development of mitigation strategies to strengthen GNSS technology, improving performance and offering more reliable GNSS globally.

1 INTRODUCING

1.1 GNSS INTERFERENCE

Global Navigation Satellite Systems (GNSS) are integral to critical infrastructure across sectors such as aviation, construction, and finance, owing to their precise positioning and timing capabilities, [1]. With increasing reliance, concerns over GNSS signal interference have grown, prompting warnings from the European Union Aviation Safety Agency (EASA) regarding associated aviation safety risks, [2]. Aircraft manufacturers, including Airbus, actively monitor and investigate GNSS interference to mitigate its impact, identifying key sources of disruption originating from personal electronic devices, tv broadcast station malfunctions, and military jamming [3].



Figure 1: Snippet of ADS-B GNSS interference heatmap on 25th April 2025. Red, Yellow and Green areas represent NIC ranges of 0%-5%, 5%-20%, and 20%+, [4].

Liu [5], describes research by Stanford University GPS Laboratory which utilized Automatic Dependent Surveillance-Broadcast (ADS-B) data to identify global GNSS interference hotspots. ADS-B transmits aircraft GNSS data to ground stations for relay to Air Traffic Control (ATC) and is widely used in European and American Aviation. This data enables the generation of interference heatmaps (Figure 1) by recording the Navigation Integrity Category (NIC), which measures the ratio of GNSS position error to a defined containment radius¹. As Trýb notes in [6], such heatmaps are vital for guiding mitigation strategies, such as designing system redundancies and enhancing operator training. However,

ADS-B coverage is limited, with dense monitoring in Europe and America, contrasted by sparse data in areas such as Africa and Asia, (as shown in Figure 1). This disparity arises due to aircraft-based systems being constrained by commercial aviation routes. A space-based system offers a promising solution by providing global coverage that is independent of commercial aviation routes.

1.2 JAMSAIL MISSION

The University of Nottingham space research group NottSpace is developing JamSail, a 3U CubeSat [7], shown in Figure 2, to study GNSS interference from space, with a secondary mission to perform a technology demonstration of a Solar Sail, [8]. A detailed mission overview is provided by Cormier in [8], including JamSail's orbital parameters. JamSail's mission profile is illustrated in Figure 3. The main modes of interest for this paper are: detumbling, & attitude determination (T3) and nominal operations (T6). In the detumbling phase, JamSail must reduce its angular velocity to maintain stability. During nominal operations, the X+ axis must consistently point nadir.

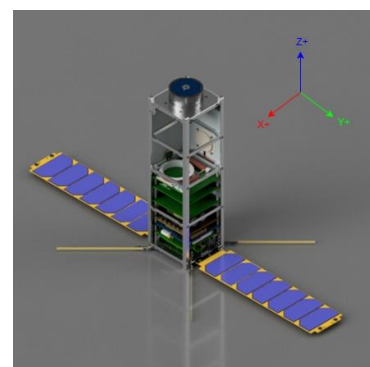


Figure 2: CAD Rendering of JamSail CubeSat with x+ and y+ Solar panels, and GNSS payload removed. Axis system represents the orientation of JamSail's body frame.

¹ For ADS-B system, the containment radius is 370.4m as defined in Title 14 of Code of Federal Regulations (CFR) Part 91 [6]

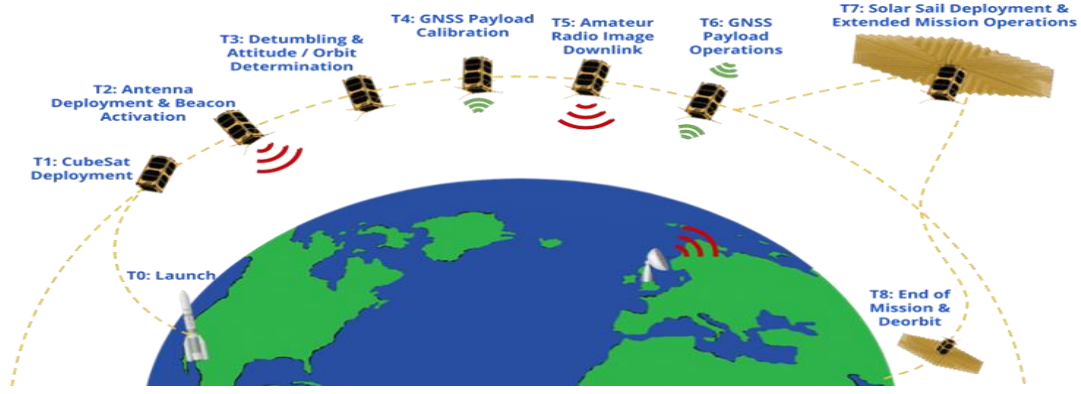


Figure 3: JamSail Mission Profile from Launch (T0), through Start Up (T1-5), to Mission Operations (T6-7) to Deorbiting (T8), taken from [8].

A satellite's attitude describes its orientation relative to an inertial reference frame. The Attitude Determination and Control System (ADCS) is responsible for estimating and controlling this orientation, ensuring mission objectives are achieved. The role of the ADCS in JamSail is to perform nadir pointing (earth pointing) orientation of the X^+ axis. JamSail will be in a Sun-synchronous orbit, with orbital parameters described in [8].

1.3 AIMS & OBJECTIVES

The project's aim is to design a CubeSat to detect and characterize GNSS interference around the globe. To support this mission, the ADCS is developed to enable precise payload operation. The system objectives, outlined in Table 1, define the requirements necessary to achieve this goal. This paper details the ADCS design and its verification and validation against the mission requirements.

Table 1: Table containing ADCS requirements for JamSail

Req-Id	Description
R-ADCS-1	The ADCS shall provide pointing accuracy within 20° in each axis.
R-ADCS-2	The ADCS shall provide the pointing accuracy defined in R-ADCS-1 for a minimum of 10 minutes at a time.
R-ADCS-3	The ADCS shall track the earth and not have pointing accuracy drift greater than $1.0^\circ \cdot s^{-1}$
R-ADCS-4	The ADCS shall detumble from a rotation rate of $90^\circ \cdot s^{-1}$ to less than $5^\circ \cdot s^{-1}$ in 60s.
R-ADCS-5	The ADCS shall have a determination accuracy of 10°

Given the challenges of replicating the space environment on Earth, with ADCS development relies primarily on high-fidelity simulators for performance & functional V&V [9, 10]. In accordance with ECSS-E-ST-60-30C guidelines [11], a comprehensive simulation framework was implemented to model all required ADCS operational modes and functions. This is in line with the verification methodology outlined in ECSS standard 5.4.4 from [11]. The development process, shown in Figure 4, involves algorithm V&V within the simulation environment prior to integration with JamSail's On-Board Computer (OBC). This paper focuses exclusively on the simulation-based V&V of the ADCS design. Hardware implementation is beyond the current scope and will be addressed in future phases of the JamSail mission.

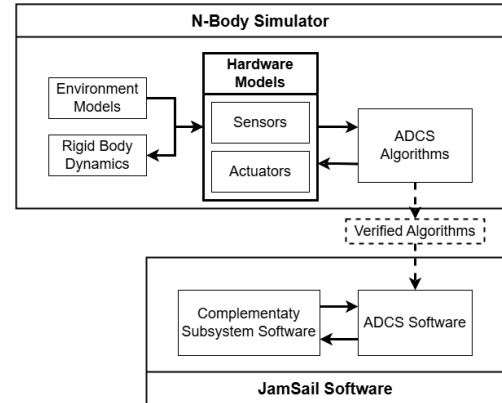


Figure 4: JamSail ADCS verification, validation and integration methodology.

2 METHODOLOGY

2.1 N-BODY SIMULATOR

The N-Body Simulator (NBS) is a modular simulator developed as part of this project in accordance with ECSS-E-TM-40-07, [12]. As highlighted previously, its main purpose is for ADCS V&V through dynamic simulation of multiple rigid bodies, including satellites and celestial bodies, alongside sensors, actuators, and environmental effects. Interfaces between models are defined with a hierarchical architecture structure, in accordance with standard 6.4.2 from [12], allowing for future development of packages to enhance the simulators capabilities. Each model within the simulator was tested in isolation, as in accordance with standard 6.6.1.7 in [12]. A full description of the NBS is beyond the scope of this paper, however, key models are discussed below.

IGRF Model

The International Geomagnetic Reference Field (IGRF) is a standard model which models the earth's magnetic field using spherical harmonics, [13]. Navabi in [14] describes how to implement the IGRF model and its variations. For simulating Earth's magnetic field, an order 13 IGRF spherical harmonic was used. The IGRF model can also be used for estimating the magnetic field in the fixed frame, as required by EKF. While higher order implementations offer more accurate estimates, they come at the cost of greater processing power. Due to the orbit of JamSail getting close to the poles, a low order IGRF model will not have the accuracy required to give reliable estimates of the magnetic field. Hence, a 3rd order IGRF model, also known as an Octupole model, was used.

Sensor & Actuator Models

Sensors are modelled by adding the true value and a zero bias gaussian noise, $\sim N(0, \sigma)$, shown by Eq.(16). $v(t) \sim N(0, R)$ describes the gaussian noise distribution from the sensors, with the sensor noise covariance, R , is such that $R = \text{diag}(\sigma_\omega, \sigma_B, \sigma_s)$. Here σ_i is the standard deviation for the gaussian noise of sensor i . An experiment was conducted on the Abacus to characterize the noise of the sensors. The test set up is shown in Figure 5, with results shown in Table 2.

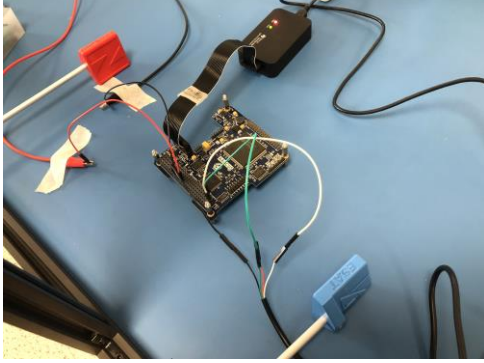


Figure 5: Sensor Characterisation set up. 50,000nT magnetic field was applied along each magnetometer axis as a controlled baseline.

Table 2: Noise standard deviation from Abacus sensors.

Sensor Noise	Value
Magnetometer Noise, σ_B (nT)	650
Gyro Noise, σ_ω (rad · s ⁻¹)	0.001

The SS200 sun sensor was selected due to its compact size, high fov, and high accuracy, [15]. From the datasheet, the noise distribution is $\sigma_s = 0.03^\circ$.

A magnetorquer is employed as the primary actuator for controlling JamSail's attitude. The iMTQ, [16], was selected due to its established space heritage and the limited availability of suitable commercial off-the-shelf (COTS) alternatives. The torque generated by a magnetorquer is given by Eq.(1), [17].

$$\vec{\tau} = \vec{M} \times \vec{B} \quad (1)$$

Here, $\vec{M}$, is the dipole and $\vec{B}$ is the local magnetic field. For a single axis, the dipole moment is calculated as, [17]:

$$M_i = N_i A_i i_i \quad (2)$$

with N_i representing the number of coil turns, A_i the loop area, and i_i the control current. To achieve a desired control torque, $\vec{\tau}_c$, the required dipole is determined by:

$$\vec{M} = \frac{\vec{B} \times \vec{\tau}_c}{\|\vec{B}\|^2} \quad (3)$$

In simulation, the magnetometer measurements are used to estimate the dipole required to generate the control torque. However, the actual magnetic field is used to simulate the resulting torque, providing a more realistic assessment of control performance under field measurement uncertainties.

2.2 KEPLERIAN PROPAGATION & VERIFICATION OF NBS DYNAMICS

Keplerian propagation models the position and velocity of orbiting bodies using classical orbital elements. An in-depth treatment is provided by Wie [17]. Within the simulator, Keplerian propagation was employed both to initialize orbital conditions and to assess the numerical accuracy of the NBS. By comparing the analytically predicted Keplerian trajectories of JamSail and the Sun relative to Earth over a single orbit with the simulator's outputs, position errors were quantified and shown in Table 3.

Table 3: Table showing error between keplerian position and NBS simulated position over a 2 orbits of JamSail (~194 mins).

Parameter	Value (km)	Percentage of Semi Major Axis ² (%)
Earth Mean Error	0.11	7.89×10^{-8}
Earth RMS Error	0.13	9.09×10^{-8}
JamSail Mean Error	9.096	0.13
JamSail RMS Error	12.9	0.18

The larger errors observed in JamSail result from simulating gravity effects caused by both sun and earth, while Kepler's method only models a two-body system. The Earth-Sun system shows smaller discrepancies due to JamSail's relative negligible mass. Accuracy could be improved by using higher-order integration methods, such as Runge-Kutta, [17], however, as the error as percentage of semi major axis is very small, these errors were deemed acceptable. Operationally, Keplerian propagation is employed by the ADCS to estimate JamSail and Sun positions in the Earth-Centered Inertial (ECI) frame. These estimates support Sun vector computation and IGRF-based magnetic field determination. Orbital elements will be updated on each ground pass to maintain estimation accuracy throughout the mission.

2.3 QUATERNIONS

JamSail's attitude is represented by a unit quaternion, $\vec{q}$, where $\|\vec{q}\| = 1$. This quaternion comprises the Gibbs vector, $\vec{g}$, and scalar component, q_s , bound such that $0 \leq q_s \leq 1$, [18] as defined in Eq.(4).

$$\vec{q} = [q_x \ q_y \ q_z \ q_s]^T = [\vec{g} \ q_s]^T \quad (4)$$

An alternative quaternion representation uses a rotation axis, $\hat{n}$, and angle, θ , with $0^\circ \leq \theta \leq 180^\circ$, as shown in Eq.(5).

$$\vec{q} = \left[\sin\left(\frac{\theta}{2}\right) \hat{n} \ \cos\left(\frac{\theta}{2}\right) \right]^T \quad (5)$$

A vector in the fixed frame, $\vec{r}_{Fix}$, can be rotated to the body frame, $\vec{r}_{bod}$, using the quaternion rotation, defined in Eq.(6). Here, $\vec{q}^*$ is the quaternion conjugate, defined in Eq.(7), and $\otimes$ is a quaternion multiplication operator, defined in Eq.(8), [18].

$$\vec{r}_{bod} = \vec{q}_{Fix \rightarrow Bod} \otimes \vec{r}_{Fix} \otimes \vec{q}_{Fix \rightarrow Bod}^* \quad (6)$$

²Semi major axis of earth and JamSail, is 149.598×10^6 km, [24], and 7000 km respectively, [9].

$$\vec{q}_{Fix \rightarrow bod}^* = \vec{q}_{bod \rightarrow Fix} = [-q_x \ -q_y \ -q_z \ q_s]^T \quad (7)$$

$$\vec{q}_1 \otimes \vec{q}_2 = \begin{bmatrix} q_{s-1}\vec{g}_2 + q_{s-2}\vec{g}_1 + \vec{g}_1 \times \vec{g}_2 \\ q_{s-1}q_{s-2} - \vec{g}_1 \cdot \vec{g}_2 \end{bmatrix} \quad (8)$$

Given an angular rate, $\vec{\omega}$, the derivative of the quaternion can be found as such, [18].

$$\dot{\vec{q}} = \frac{1}{2} \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \vec{q} = \frac{1}{2} \Omega \vec{q} \quad (9)$$

$$\dot{\vec{q}} = \frac{1}{2} \begin{bmatrix} q_s & -q_z & q_y \\ q_z & q_s & -q_x \\ -q_y & q_x & q_s \\ -q_x & -q_y & -q_z \end{bmatrix} \vec{\omega} = \frac{1}{2} Q \vec{\omega} \quad (10)$$

The rotation can be equivalently expressed using a Direction Cosine Matrix (DCM), A , as described in Eq. (13), [18].

2.4 FIX & BODY FRAME

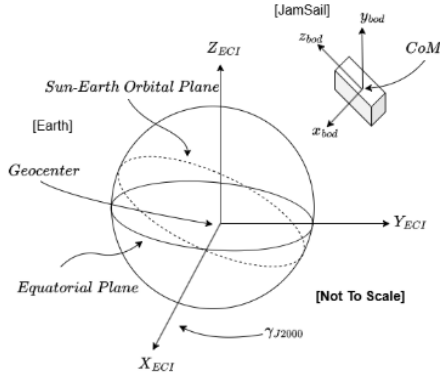


Figure 6: Diagram showcasing nadir pointing body frame of JamSail relative to fixed ECI frame.

In this paper, the fixed reference frame is the Earth-Centered Inertial (ECI) frame, defined with its origin at Earth's center of mass (Geocenter). The z^+ axis aligns with the rotation vector of earth, the x^+ axis points towards the mean Vernal equinox of epoch J2000, γ_{J2000} , and the y^+ axis completes a right-handed orthogonal system. For the JamSail body frame, the origin is located at the CubeSats CoM, with its x^+ axis aligned with the payload pointing direction, the z^+ axis defines the longitudinal axis, and y^+ axis completes a right-handed orthogonal system.

2.5 RIGID BODY DYNAMICS

When it comes to simulating the rigid body dynamics of JamSail, Eq.(11-12) was used. I and m are the inertia matrix and mass of JamSail respectively. Further, $\vec{\tau}_N$ and $\vec{F}_N$ are the total torques, and forces applied to JamSail. For current simulations, only the gravity force and magnetorquer input torques are simulated. External loads such as drag forces, and gravity torques are left to be implemented in future iterations of the NBS.

$$\ddot{\vec{r}} = \frac{\vec{F}_N}{m} \quad (11)$$

$$\dot{\vec{\omega}} = I^{-1}(\vec{\tau}_N - \vec{\omega} \times I \vec{\omega}) \quad (12)$$

The inertia matrix and mass of JamSail were estimated from hand calculations and a CAD model to be:

$$I = \begin{bmatrix} 0.0181640 & -0.0000022 & -0.0001082 \\ -0.0000022 & 0.0182437 & -0.0000709 \\ -0.0001082 & -0.0000709 & 0.0033638 \end{bmatrix} \text{kgm}^2$$

$m = 4\text{kg}$

2.6 ADCS ARCHITECTURE

JamSail's ADCS state machine is illustrated in Figure 7. The system transitions through five operational modes based on mission conditions and sensor inputs.

Start-Up Mode:

Upon power-up or reboot, JamSail enters Start-Up mode, during which a timer is initialized. Once this timer reaches 600s, the state will update to Sun Searching mode. If the angular rates are greater than $20^\circ \cdot s^{-1}$, the state updates to Detumbling mode. During start up mode, the detumbling control algorithm is used.

Sun Searching Mode:

In this mode, JamSail rotates about its x^+ -axis at a rate of $2^\circ \cdot s^{-1}$ to locate the sun. It uses data from its gyroscope, as attitude cannot be reliably estimated at this point. $2^\circ \cdot s^{-1}$ was chosen because it equates to roughly 32 revolutions per orbit while keeping the angular rate low that control of the CubeSat will not be lost.

Sun Tracking Mode:

On detection of a valid sun signal, the ADCS aligns the z^+ axis towards the sun. As the sun sensor is pointing along the z^+ axis, this allows for rapid convergence of attitude estimation, with the added bonus of having maximum solar power generation, due to the configuration of the solar panels as shown in Figure 2. Once the sun has been tracked continuously for 600s, the system transitions to Nominal Operation.

Nominal Operation:

The ADCS will reorient JamSail such that x^+ -axis points nadir, while the z^+ axis tracks the sun within the plane formed by the sun and nadir vectors. If the sun visibility is lost for 300s (e.g. during eclipse), the system reverts to sun searching mode. This 300s buffer is to give the system a chance to recover.

Detumbling Mode

If at any point, excluding Start-Up, the magnitude of $\vec{\omega}$ exceeds $20^\circ \cdot s^{-1}$, Detumbling mode will be activated. This rate was chosen as the maximum controlled rate expected is $10^\circ \cdot s^{-1}$. Then, the control algorithm will reduce the angular velocities until the magnitude reaches $5 \text{ deg} \cdot s^{-1}$, before returning the state to *Sun-Searching* mode. This way, if there is an unknown disturbance, JamSail will be able to recover and continue with its operations.

$$\vec{r}_{bod} = A \vec{r}_{Fix} = \begin{bmatrix} q_x^2 - q_y^2 - q_z^2 + q_s^2 & 2(q_x q_y + q_z q_s) & 2(q_x q_z - q_y q_s) \\ 2(q_x q_y - q_z q_s) & -q_x^2 + q_y^2 - q_z^2 + q_s^2 & 2(q_y q_z + q_x q_s) \\ 2(q_x q_z + q_y q_s) & 2(q_y q_z - q_x q_s) & -q_x^2 - q_y^2 + q_z^2 + q_s^2 \end{bmatrix} \vec{r}_{Fix} \quad (13)$$

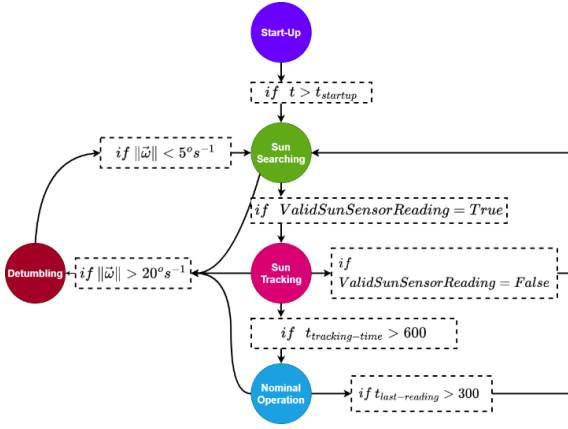


Figure 7: JamSail's ADCS State Machine system. Logic starts in start-up mode, and moves in direction of arrows, once conditions in boxes are met.

2.7 ATTITUDE DETERMINATION

Due to budgetary constraints, JamSail's ADCS was designed around the available hardware, specifically the ABACUS OBC by GAUSS Srl, [19], which includes a magnetometer and gyroscope. As the University of Nottingham's first CubeSat mission, the design emphasized reliability and modularity to support future ADCS development.

The sensor suite (magnetometer, gyroscope, and sun sensor) was selected based on successful implementations in missions such as M-Sat [20, 21], UVSQ-SAT [22], and IlliniSat [23]. These missions were chosen for their comparable scale, sensor constraints, and demonstrated use of low-cost, reliable ADCS architectures—making them directly relevant to JamSail's design objectives. These studies also demonstrated the efficacy of Extended Kalman Filters (EKFs) for attitude estimation. While M-Sat relied solely on a magnetometer with stochastic modeling, UVSQ-SAT and IlliniSat achieved higher reliability using multi-sensor EKF approaches. While the referenced missions utilized combinations of magnetometers, gyroscopes, and sun sensors, none incorporated all three simultaneously. This study builds on their foundations by evaluating the performance of fusing all three sensor modalities within a unified EKF architecture. Further, this paper extends their work to incorporate control algorithms.

Extended Kalman Filter

To estimate JamSail's attitude and angular velocity, an EKF is employed, expanded from [20-23], using the state vector:

$$\vec{x}(t) = [\vec{q}(t)^T \quad \vec{\omega}(t)^T]^T \quad (14)$$

The system dynamics are defined by:

$$\dot{\vec{x}}(t) = f(\vec{x}(t), \vec{u}(t)) + \vec{w}(t), \quad \vec{w}(t) \sim N(0, Q) \quad (15)$$

The measurement vector comprises angular velocity, magnetic field, and sun vector data:

$$\vec{y}(t) = \begin{bmatrix} \vec{\omega}(t) \\ \vec{B}(t) \\ \vec{s}(t) \end{bmatrix} + \vec{v}(t), \quad \vec{v}(t) \sim N(0, R) \quad (16)$$

The Jacobian of the state dynamics, $F(t)$, are derived accordingly:

$$F(t) = \frac{\delta f(\vec{x}(t))}{\delta \vec{x}} = \begin{bmatrix} \frac{1}{2} \Omega(t) & \frac{1}{2} Q(t) \\ 0_{3 \times 4} & \Pi(t) \end{bmatrix} \quad (17)$$

Where Π can be found using Eq.(18):

$$\Pi = \frac{\delta \dot{\vec{\omega}}}{\delta \vec{\omega}} = \begin{bmatrix} 0 & \frac{I_{yy} - I_{zz}}{I_{xx}} \omega_z & \frac{I_{yy} - I_{zz}}{I_{xx}} \omega_y \\ \frac{I_{zz} - I_{xx}}{I_{yy}} \omega_z & 0 & \frac{I_{zz} - I_{xx}}{I_{yy}} \omega_x \\ \frac{I_{xx} - I_{yy}}{I_{zz}} \omega_y & \frac{I_{xx} - I_{yy}}{I_{zz}} \omega_x & 0 \end{bmatrix} \quad (18)$$

The measurement model is defined as:

$$h(\vec{x}) = \begin{bmatrix} \vec{\omega}^* \\ A \hat{B}_{Fix}^* \\ A \hat{s}_{Fix}^* \end{bmatrix} \quad (19)$$

Where A is as defined in Eq.(13). The asterisk denotes that the quantity is an estimate. The measurement Jacobian is defined as such:

$$H(t) = \frac{\delta h(\vec{x}(t))}{\delta \vec{x}} = \begin{bmatrix} \mathbb{I}_{3 \times 3} & 0_{3 \times 4} \\ 0_{3 \times 3} & \Theta_B(\vec{q}) \\ 0_{3 \times 3} & \Theta_s(\vec{q}) \end{bmatrix} \quad (20)$$

where $\mathbb{I}$ is an identity matrix and,

$$\Theta_B(\vec{q}) = \begin{bmatrix} \frac{\delta A_{Fix \rightarrow bod}}{\delta q_x} \hat{B}_{Fix}^* & \dots & \frac{\delta A_{Fix \rightarrow bod}}{\delta q_s} \hat{B}_{Fix}^* \end{bmatrix} \quad (21)$$

$$\Theta_s(\vec{q}) = \begin{bmatrix} \frac{\delta A_{Fix \rightarrow bod}}{\delta q_x} \hat{s}_{Fix}^* & \dots & \frac{\delta A_{Fix \rightarrow bod}}{\delta q_s} \hat{s}_{Fix}^* \end{bmatrix} \quad (22)$$

The partial derivatives are derived from Eq.(9), with Babcock providing an example in [23]. The EKF operates by initializing the state and covariance, P :

$$\vec{x}(0) = [0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0]^T, \quad P(0) = \mathbb{I}_{7 \times 7} \quad (23)$$

At each step, k , the algorithm proceeds as follows:

1. Update System Variables using current measurements and estimate of dynamics.

2. Compute Kalman Gain. K

$$K(k) = P(k)H^T(k)(H(k)P(k)H^T(k) + R)^{-1} \quad (24)$$

3. Update State & Error Covariance

$$\vec{x}^+(k) = \vec{x}(k) + K(k)[h(\vec{x}(k)) - \vec{y}(k)] \quad (25)$$

$$P^+(k) = (\mathbb{I}_{7 \times 7} - K(k)H(k))P(k) \quad (26)$$

4. Find control torques, and estimate applied torques using Eq.(3).

5. Compute derivatives and integrate the state and error covariance

$$\dot{P}(k) = P^+(k)F(k) + F^T(k)P(k) + Q \quad (27)$$

2.8 ATTITUDE CONTROL

Finding Required Orientation

The required quaternion, $\vec{q}_r$, can be calculated using:

$$\vec{q}_r = \vec{q}_{primary} \otimes \vec{q}_{secondary} \quad (28)$$

The primary required attitude, $\vec{q}_{primary}$, is defined by aligning the CubeSat's x^+ body axis with the nadir vector, which can be calculated using Keplerian propagation as $\vec{r}_{JamSail \rightarrow o} = -\vec{r}_{o \rightarrow JamSail}$, where o is the origin of the fixed frame. The secondary required attitude, $\vec{q}_{secondary}$, is constructed to align the z^+ body axis in the plane made with the sun vector and nadir vector.

Eq.(29) & Eq.(30) can be used to find the rotation axis and rotation angle, before being substituted into Eq.(5) to find the required angle.

$$\hat{n} = \frac{\vec{r}_{axis} \times \vec{r}_{direction}}{\|\vec{r}_{axis} \times \vec{r}_{direction}\|} \quad (29)$$

$$\theta = \cos^{-1} \left(\frac{\vec{r}_{axis} \cdot \vec{r}_{direction}}{\|\vec{r}_{axis}\| \|\vec{r}_{direction}\|} \right) \quad (30)$$

For $\vec{q}_{primary}$, $\vec{r}_{axis}$ is equal to $x^+ = [1, 0, 0]^T$, and $\vec{r}_{direction}$ is equal to $\vec{r}_{JamSail \rightarrow o}$. To find $\vec{q}_{r-secondary}$, the rotation axis is the nadir vector, which, after the primary rotation, is x^+ axis. Then, to find the rotation angle, Eq.(31) can be used.

$$\theta_{secondary} = \tan^{-1} \left(\frac{-S_{y-bod}}{S_{z-bod}} \right) \quad (31)$$

Nominal Operations Control

During nominal operation, quaternion-based rotation logic is employed, [17]. the error quaternion is first calculated using Eq.(32).

$$\vec{q}_e = \begin{bmatrix} q_{r-s} & q_{r-z} & -q_{r-y} & -q_{r-x} \\ -q_{r-z} & q_{r-s} & q_{r-x} & -q_{r-y} \\ q_{r-y} & -q_{r-x} & q_{r-s} & -q_{r-z} \\ q_{r-x} & q_{r-y} & q_{r-z} & q_{r-s} \end{bmatrix} \vec{q} \quad (32)$$

Following this, a quaternion feedback controller is used based on $\vec{q}_e$ Gibbs vector, shown below:

$$\vec{\tau}_c = -K\vec{q}_e - C\vec{\omega} + \vec{\omega} \times I\vec{\omega} \quad (33)$$

K and C are diagonal gain matrices. This formulation allows smooth convergence by compensating for attitude error, damping rotational rates, and accounting for cross-coupling dynamics, [17]. For nominal operation, the control coefficients where set to be $K = \text{diag}(0.06, 0.06, 0.01)$, $C = \text{diag}(0.06, 0.06, 0.01)$.

Sun Searching & Detumbling Operations Control

For sun searching and detumbling, the controller used is based around controlling $\vec{\omega}$, which is shown in Eq.(34).

$$\vec{\tau}_c = C(\vec{\omega}_r - \vec{\omega}) + \vec{\omega} \times I\vec{\omega} \quad (34)$$

When the ADCS is performing Sun Searching, the required angular rate is $\vec{\omega}_r = [2.0 \ 0.0 \ 0.0]^T \cdot s^{-1}$, with control coefficients of $C = \text{diag}(0.06, 0.06, 0.01)$. For detumbling $\vec{\omega}_r = [0.0 \ 0.0 \ 0.0]^T \cdot s^{-1}$ with control coefficients $C = \text{diag}(0.6, 0.6, 0.1)$.

Sun Tracking Operations Control

Once the sun is detected, the angles around the body x and y axis are found as follows

$$\theta_x = \tan^{-1} \left(\frac{S_{y-bod}}{S_{z-bod}} \right) \quad (35)$$

$$\theta_y = \tan^{-1} \left(\frac{S_{x-bod}}{S_{z-bod}} \right) \quad (36)$$

The components of the sun vector in the body frame can be taken directly from the sun sensor, hence no knowledge of the attitude is required. The control logic is then described in Eq.(37).

$$\vec{\tau}_c = K\theta_e - C\vec{\omega} + \vec{\omega} \times I\vec{\omega} \quad (37)$$

Where $\theta_e = [\theta_x \ \theta_y \ 0]^T$. The coefficients for K and C are the same as in the nominal mode.

2.9 SIMULATION TEST CASES

To validate the ADCS design, 3 test cases were simulated to establish how the system will perform under different conditions.

Nominal Operation Test Case

This test case establishes how the system would perform in an ideal environment. It is simulated for two periods of orbit ($\sim 194\text{mins}$), so that the performance to see if the system can perform continuously. This test case is intended to validate that the system meets requirements 1, 2, 3, & 5. It is assumed that if the system can perform over two orbits that it will be able to perform continuously to perform its mission. This is due to how the ADCS logic system follows a cyclical transition of operations for conditions expected to see during orbit. The main area being assessed is the systems ability to perform its mission over multiple orbits, assuming that if it can perform over two, in a stable control manner it will continue to keep performing for multiple orbits.

Initial conditions are presented in [8], where the quaternion represents a 321 Euler rotation of 90° , 0° and 180° around X, Y, & Z respectively. This introduces a determination error of 90° , 90° & 180° between the estimated and actual X, Y & Z axis respectively.

Table 4: Initial conditions for test case 1.

Initial Condition	Value
Quaternion, $\vec{q}_o$ (-)	$[0.0 \ 0.7071068 \ 0.7071068 \ 0.0]^T$
Angular Rate, $\vec{\omega}_o$ ($^\circ \cdot s^{-1}$)	$[0.0 \ 0.0 \ 0.0]^T$

Detumbling Test Case

During this test case, multiple test runs are simulated to represent worst case launcher separation, (in line with standard 5.3.1 in [11]). This is to show that the control system can detumble the CubeSat and show that requirement 4 is met.

The test runs, summarized in Table 5, are simulated for 60s. Test Run 1 isolates single-axis rotation to establish baseline detumbling behavior. Test Run 2 has a magnitude of $90^\circ \cdot s^{-1}$, and evaluates performance under coupled multi-axis motion. This demonstrates robustness and compliance with requirement R-ADCS-4, which mandates

recovery from rates up to $90^\circ \cdot s^{-1}$. Test Run 3 represents a worst-case scenario, applying high angular velocities across all axes to assess system recovery from extreme initial conditions. All test runs have same initial quaternion as defined in Table 4. All test runs used the same initial quaternion as in Nominal Operation Test Case.

Table 5: Initial Angular rates for test runs to simulate detumbling algorithm.

Test Run	Initial Angular Rate, $\vec{\omega}_0$ ($^\circ \cdot s^{-1}$)
Test Run 1	$[0.0 \ 90.0 \ 0.0]^T$
Test Run 2	$[20.0 \ 78.0 \ 40.0]^T$
Test Run 3	$[90.0 \ 90.0 \ 90.0]^T$

Disturbance Recovery Test Case

This test case serves to validate the system against all requirements and show the robustness of the system. In this test case, it will start with high angular rates and be simulated for a single orbit ($\sim 97mins$). Further, there will be disturbance torque off $0.001N \cdot m$ applied to all three axes for 300s beginning at 2,500s. This is to see how the system will perform when a disturbance is applied to the system. This test case will have the same initial conditions from Test Run 3 from Table 5.

Atmospheric and gravity disturbance torques, are expected to be in the order of $10 \mu N \cdot m$, [10]. Hence this test will show how the system handles external disturbances, not accounted for in the simulation, validating robustness of the design.

The main area being assessed by this test case is the ability of the system to handle extreme conditions and still collect value data to perform its mission. Further, it will also show the system's ability to recover from adverse effects, giving high confidence that it will be able to handle situations not accounted for in simulation.

3 SIMULATION RESULTS & DISCUSSION

3.1 NOMINAL OPERATION TEST CASE

Figure 9(a) shows that during nominal mode, attitude determination error drops below 10° , and remains steadily at 5° for 2,355s ($\sim 39mins$), demonstrating compliance with Requirement R-ADCS-5. Transitions to sun-search from nominal mode coincide with eclipse periods, indicating that the control logic tracks the sun to maintain attitude accuracy for maximum period of time.

A transient increase in error occurs near 9,700s, is due to sun sensor dropout from JamSail being in eclipse. Further, it can be seen that during portions of the eclipse the ADCS is not able to meet requirement R-ADCS-5. A minimum of 2 vectors is required to determine an orientation with 3 degrees of freedom, [10]. While tracking JamSail's angular rates and numerically integrating can estimate how the orientation changes with time, this is susceptible to numerical drift, [10]. For future system designs, the integration of more sensors can be used to improve the ADCS performance, extending reliable attitude

estimation in eclipse portions of orbit. This process will involve simply updating the measurement jacobian (shown in Eq.(20)). With their being some indication that the attitude determination accuracy could be acceptable during eclipse, payload operations could potentially still run, under the pretense understanding of the potential unreliability of the data.

Figure 9(b) illustrates a pointing accuracy of 7° at worst case during nominal period ($\sim 39mins$). This satisfies requirements R-ADCS-1 through R-ADCS-3, and ensures the GNSS payload can reliably associate interference measurements recorded with the correct regions on the globe. Importantly, these results include errors from Keplerian-based position estimates, underscoring the system's robustness to internal uncertainty. Pointing accuracy is not calculated when the state is not nominal due to the required attitude not being defined.

For comparison, Figure 8 presents results from Babcock's EKF implementation [23]. Compared to the convergence of 60s in Figure 9(a), the slower convergence of 40mins is likely due to the increased noise from photodiode-based sun vector estimation. However, both EKF's do achieve accurate attitude convergence, showing consistency across systems and further validating the performance of the current system. This also highlights having a more accurate sun sensor will significantly impact convergence time of ADCS system.

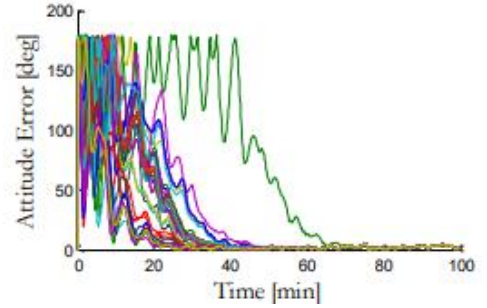


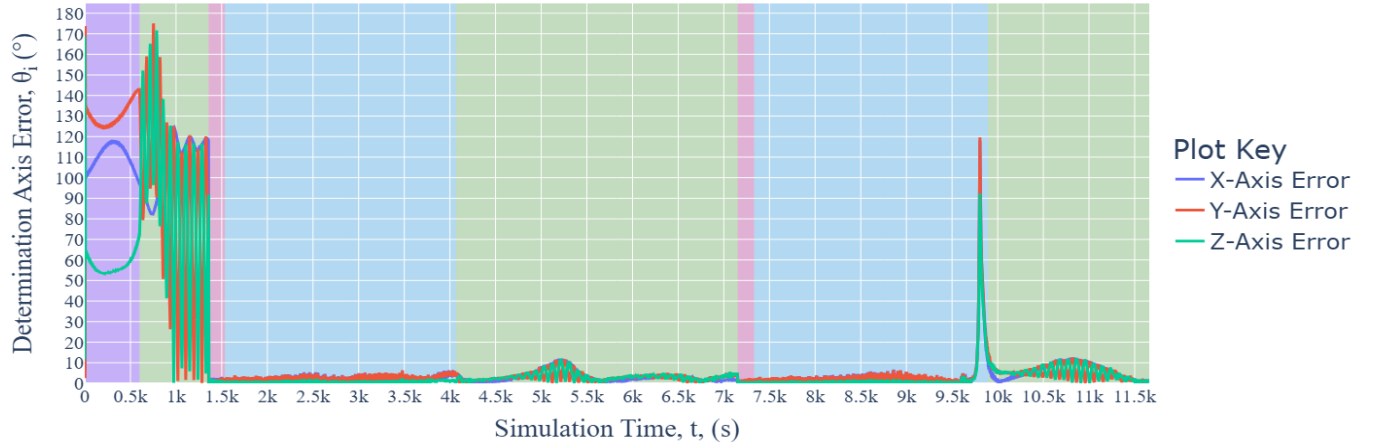
Figure 8: Convergence of EKF from 180° initial attitude error with $\sigma_s = 5^\circ$ from [23].

3.2 DETUMBLING TEST CASE

Shown in Figure 9, the ADCS successfully reduced angular rates to below $5^\circ \cdot s^{-1}$ in all cases from the initial conditions described in Table 5 while in detumbling mode, meeting requirement R-ADCS-4.

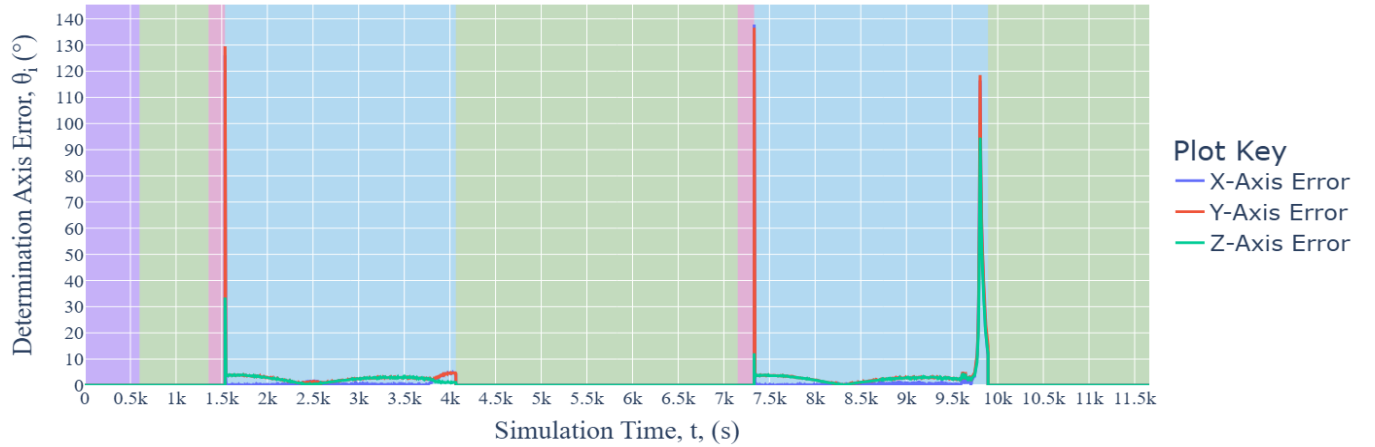
Despite successful rate reduction, the system exhibited oscillatory behavior during detumbling. This is attributed to cross-coupling in the inertia matrix (see Section 0) and the inherent limitations of magnetorquers, which cannot exert torques parallel to the magnetic field, [17]. Consequently, full three-axis decoupling remains challenging. Future work will aim to minimize cross-coupling and overshoot by optimizing the CubeSat's internal mass distribution and developing smarter control strategies that account for the rate of change in the Earth's magnetic field from the CubeSats perspective.

Determination Error Angles between True Body Axis and EKF Estimated Body Axis



(a)

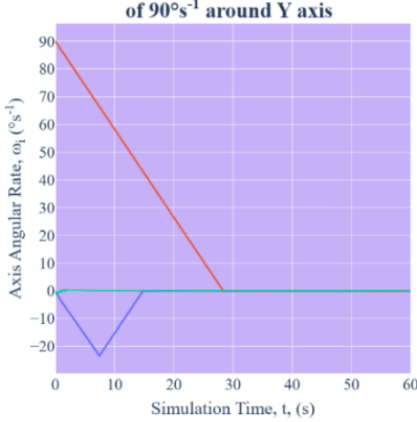
Pointing Error Angles between Commanded Axis and Actual Axis Orientation



(b)

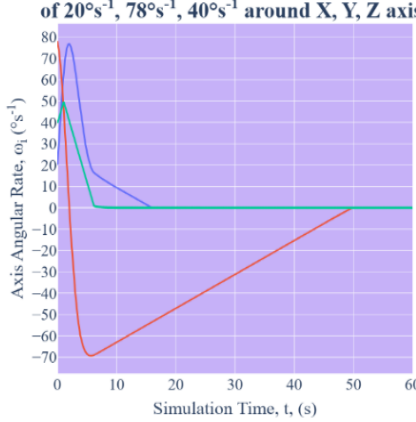
Figure 9: Determination error between the estimated and actual orientation of x^+ , y^+ and z^+ body axis (a), and the pointing error between the required and actual orientation of x^+ , y^+ and z^+ axis (b) for Nominal Operation Test Case. Background purple, green, red, and blue represent ADCS Start-Up, Sun Searching, Sun Tracking and Nominal states³ respectively.

Detumbling Performance with initial conditions of 90°s^{-1} around Y axis



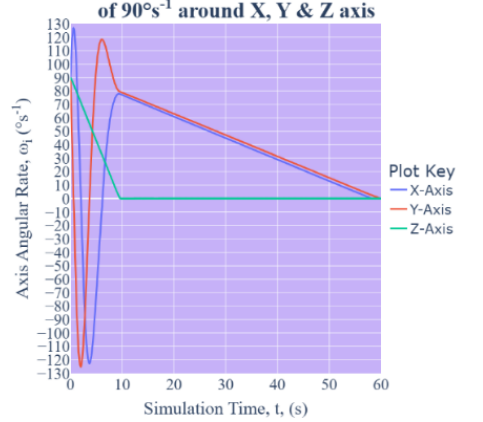
(a)

Detumbling Performance with initial conditions of 20°s^{-1} , 78°s^{-1} , 40°s^{-1} around X, Y, Z axis



(b)

Detumbling Performance with initial conditions of 90°s^{-1} around X, Y & Z axis



(c)

Figure 10: Angular rates from Detumbling Test Case with figures (a), (b) & (c) representing Test Runs 1, 2, & 3 respectively.

³ Colours reflect ADCS state colour scheme shown in Figure 7.

3.3 DISTURBANCE RECOVERY TEST CASE

For this test run, initial conditions matched those of Test Run 3 from the detumbling case to evaluate system performance under extreme conditions.

Table 6: Initial inputs for runs for Disturbance Recovery Test case.

Initial Condition	Value
Quaternion, $\vec{q}_o$ (-)	$[0.0 \ 0.7071068 \ 0.7071068 \ 0.0]^T$
Angular Rate, $\vec{\omega}_o$ ($^\circ \cdot s^{-1}$)	$[90 \ 90 \ 90]^T$

Results shown in Figure 11 demonstrate that despite the high initial angular rates, the ADCS successfully stabilizes and transitions JamSail into nominal operation. During the disturbance period, attitude determination becomes momentarily unreliable; however, the system recovers promptly, re-establishing accurate determination and control of attitude. This confirms that the design not only satisfies the requirements outlined in Table 1, but also maintains resilience under severe disturbance conditions.

Comparing Figure 11 with Figure 9, the initial high angular rates don't impact the performance of the ADCS system after it has been disturbed. The ADCS was still able to converge on an accurate attitude estimation of about 3°

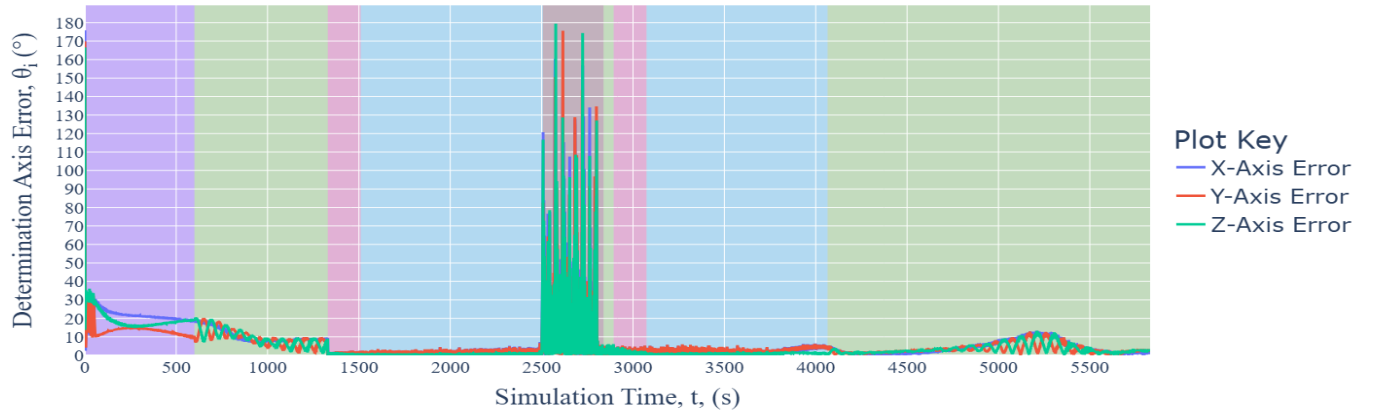
and rotate the body frame into the desired orientation to have a pointing accuracy of about 5° .

4 CONCLUSION

The aim of this project was to develop a CubeSat to detect and characterize GNSS interference around the globe. The ADCS enables operation of the payload to achieve this aim, with objectives captured as requirements for this project. The accurate characterization and geolocation would impact future GNSS technology and operations to overcome the challenge of interference, improving capabilities of these systems (like aircraft) making them safer and more reliable.

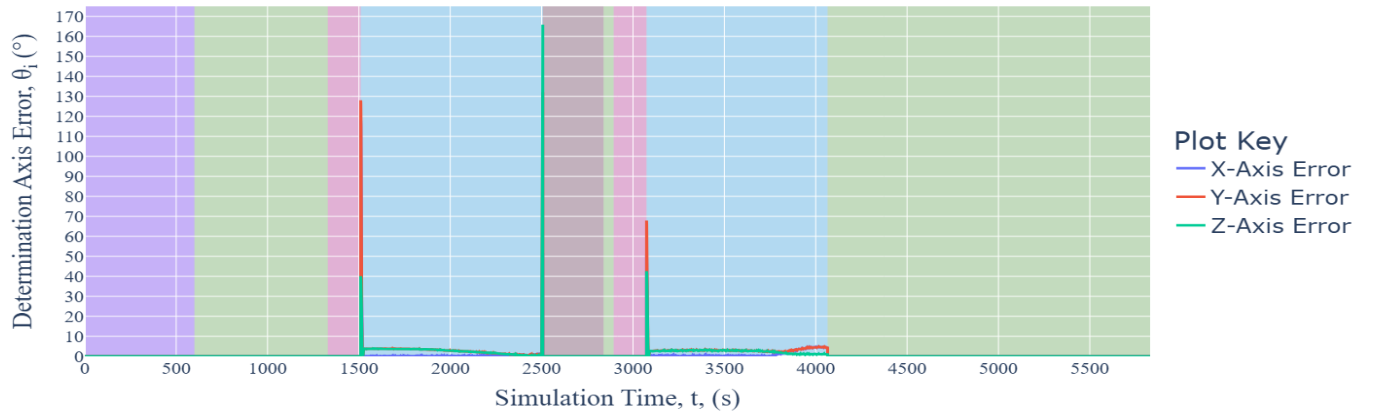
A simulation environment was developed to V&V the ADCS of JamSail. This followed ECSS standards and simulated hardware, environmental phenomena as well as ADCS algorithms. From research into the literature an EKF was designed which incorporated aspects of design from multiple references, [20-23], to include an EKF which incorporates a magnetometer, sun sensor and gyro. Further, an ADCS state machine was developed with control algorithms for each mode of operation. These were integrated alongside the determination algorithms. This way, the results shown would be representative of onboard software and how it would operate.

Determination Error Angles between True Body Axis and EKF Estimated Body Axis



(a)

Pointing Error Angles between Commanded Axis and Actual Axis Orientation



(b)

Figure 11: Figure showing the determination error between the estimated and actual orientation of x^+ , y^+ and z^+ body axis (a), and the pointing error between the required and actual orientation of x^+ , y^+ and z^+ axis (b) for Disturbance Recovery Test Case. Background purple, green, red, and blue represent ADCS Start-Up, Sun Searching, Sun Tracking and Nominal states respectively.

Several test runs were performed which indicated a determination accuracy of 5° and a pointing accuracy of 7° , which was maintained for $\sim 39\text{mins}$ per orbit. It was found that when there was an invalid sun sensor reading, the EKF would struggle to determine an accurate attitude with reliable consistency and was more susceptible to drift. This was expected to be due to the requirement of having a minimum of two vectors for attitude determination. This is an area that would be of great value to solve as it will enable operations within an eclipse portion of orbit. Potential future work is to integrate an earth sensor into the ADCS to solve this problem and enable continuous operation of payload.

Regardless, these results indicate that the ADCS can provide sufficient attitude determination accuracy to reliably geolocation GNSS interference sources, achieving the determination requirements in Table 1.

Further, the detumbling ability of the ADCS was simulated in 3 test runs. In all 3 test runs the ADCS was able to reduce the angular rates to less than $5^\circ \cdot s^{-1}$ in less than 60s. This shows that the ADCS is able to handle the extreme conditions that may occur during orbital separation.

Finally, a test case was run to see if the initial angular rates would impact nominal operation, along with if the ADCS could recover from any disturbances it may encounter. It was found that after the detumbling algorithm was performed, the ADCS system would reliably search for the sun, track upon a valid sun sensor reading, and then perform nominal operation. This shows robustness within the design.

These 3 test cases verify that the requirements of the ADCS are met. The next stage is to integrate software into JamSail's hardware. The current work shows V&V for the algorithms. Future work in expanding the capabilities of the NBS simulator to enable Hardware in the Loop testing will allow for V&V of the requirements on the flight software.

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Logbook

Meeting Minutes under 'Supervisor Meeting Minutes' section of Logbook